

Task 1: Apps and Service Models

Application/Service	Model	Reason
Instagram	SaaS	Users directly consume the completed application without managing its infrastructure.
Zomato	SaaS	Customers use a complete food-ordering application through web/mobile interfaces.
Spotify	SaaS	Users consume a complete music-streaming service without managing servers or operating systems.

Task 2: IaaS vs PaaS vs SaaS

Feature	IaaS	PaaS	SaaS
Full form	Infrastructure as a Service	Platform as a Service	Software as a Service
Provides	Virtual machines, storage, networking	Application development/deployment platform	Complete software application
User manages	OS, applications and configurations	Application/code and data	Mainly application settings/data
Scalability	High	High	Usually handled by provider
Example	AWS EC2	Google App Engine	Microsoft 365

Typical user	System administrator	Developer	End user
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Examples

IaaS — AWS EC2:

Provides virtual servers where customers can install operating systems and applications.

PaaS — Google App Engine:

Provides a managed platform for developers to deploy applications without managing the underlying server infrastructure.

SaaS — Microsoft 365:

Provides ready-to-use applications such as Word, Excel and Outlook through the Internet.

Task 3: Launch an Ubuntu EC2 Instance

Steps

1. Log in to AWS.
2. Search for EC2.
3. Open the EC2 Dashboard.
4. Select Launch instance.
5. Enter a name such as:
 - Cloud-Lab-Ubuntu
6. Select an Ubuntu Server AMI appropriate for the assignment.
7. Choose an eligible small instance type available under your account's current Free Tier terms.
8. Create/select a key pair.
9. Configure the security group carefully.
10. Allow SSH only from your own IP address where possible.
11. Launch the instance.
12. Wait until the instance reaches the Running state.

Task 4: Connect Using SSH

Linux/macOS

Example:

- `chmod 400 mykey.pem`

Then:

- `ssh -i mykey.pem ubuntu@YOUR_PUBLIC_IP`

Replace:

- `mykey.pem`

with your actual key filename and:

- `YOUR_PUBLIC_IP`

with your EC2 instance's public IP address.

Windows PowerShell

You can use the built-in OpenSSH client:

- `ssh -i .\mykey.pem ubuntu@YOUR_PUBLIC_IP`

After connecting, run:

- `uname -a`

Example Output

Your output will be different depending on the Ubuntu version/kernel:

- `Linux ip-172-31-xx-xx 6.x.x-xx-generic #xx-Ubuntu SMP ...`

Task 5: Host a Static Website on EC2

Steps

1. Connect to the EC2 instance using SSH.
2. Install a web server such as Apache or Nginx.
3. Place the HTML/CSS files in the web server's document directory.
4. Configure the EC2 security group to allow HTTP traffic on port 80.

5. Open the instance's public IP address in a web browser and verify the website.

Example Apache Installation

- `sudo apt update`
- `sudo apt install apache2 -y`